Nalla Malla Reddy Engineering College (Autonomous)

Department of Computer Science and Engineering

PRC-2

AI-Based Generative Art using Real-Time Speech Emotion and Personality Recognition

Guided By:
Sabih Ahmad Ansari
Assistant professor

Presented By:

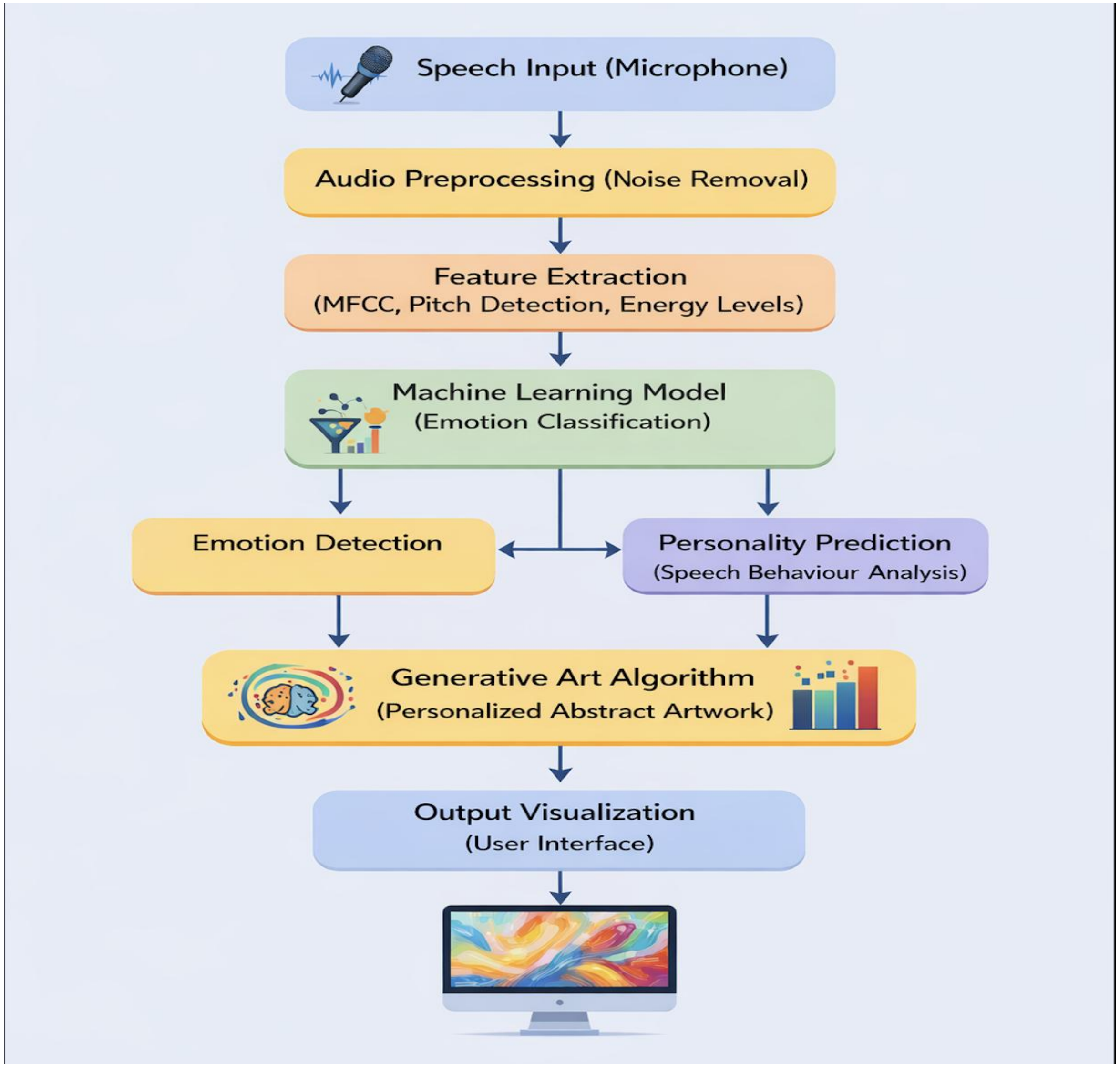
A. Ram Mohan - 23B61A0501
K. Rahul - 23B61A0525
K. Yogi Teja - 23B61A0526
M. Hari Prakash - 23B61A0535

INDEX

Abstract

This project presents a real-time speech emotion and personality recognition System for AI- Based Generative Art, an intelligent framework that transforms human voice input into expressive digital artwork. The system analyzes vocal features such as tone, pitch, intensity, and speech patterns to identify emotional states including happiness, sadness, anger, calmness, and excitement using a trained emotion recognition model. In addition to detecting emotions, the project incorporates psychometric analysis techniques to infer personality traits from speech characteristics, enabling deeper personalization of the generated artwork. By integrating speech processing, machine learning, personality assessment, and generative art techniques, the system produces meaningful emotion-driven visual content. The proposed solution can be applied in areas such as art therapy, mental health monitoring, creative design tools, personalized digital art generation, and AI-powered exhibitions, providing an innovative approach to visualizing and interpreting human emotions and personality through artificial intelligence.

Methodology



Implementation

The implementation of this project is divided into frontend and backend modules.

Frontend:

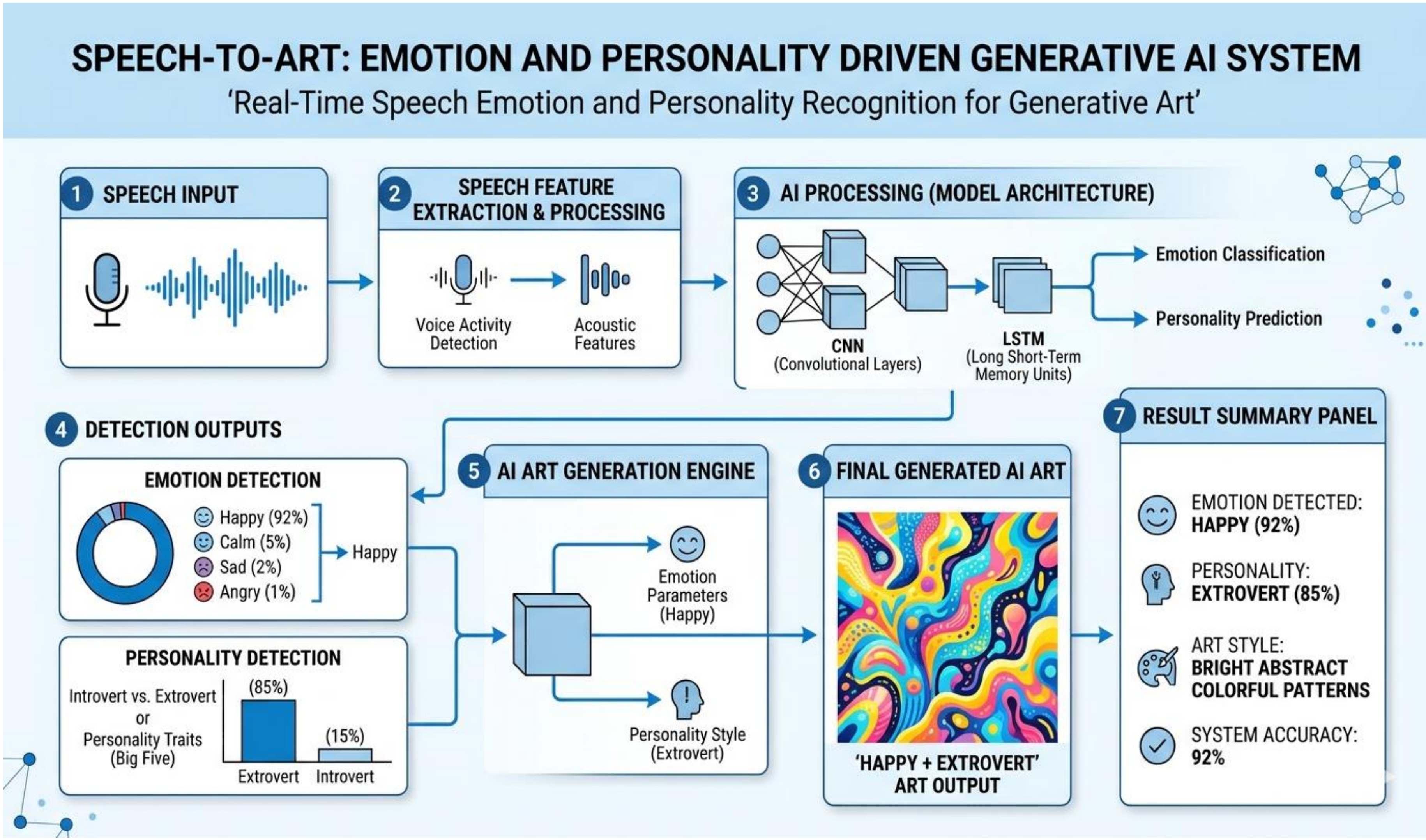
- Developed using HTML, CSS, and JavaScript
- Provides interface for voice recording
- Displays detected emotion and generated artwork
- Shows personality trait visualization

Backend:

- Implemented using Python
- Libraries used: TensorFlow, Librosa, Scikit-learn, NumPy, Matplotlib
- Audio processing for feature extraction
- Machine learning model for emotion detection
- Generative AI module for artwork creation

The system connects frontend and backend using APIs. The voice input is processed and the output is displayed as emotion results and generated images.

Final Results



Applications & Advantages

Applications

- Art therapy for emotional expression
- Mental health monitoring systems
- Entertainment industry
- Personalized AI art generation
- Human computer interaction systems
- AI creative design tools

Advantages

Advantages

- Real-time emotion detection
- Personalized artwork generation
- Combines AI and creativity
- Easy human interaction through voice
- Useful for emotional analysis
- Scalable for future AI applications

Conclusion

This project successfully demonstrates the integration of speech emotion recognition, personality analysis, and AI-based generative art. The system is capable of identifying emotional states from voice input and transforming them into meaningful visual representations. The project shows how artificial intelligence can be used not only for analysis but also for creative applications. The developed system provides a new way to visualize human emotions using technology and can be extended for various real-world applications.

Future Scope

This project can be further improved by increasing the dataset size to improve emotion detection accuracy. Deep learning models such as transformers can be used for better prediction. Real-time mobile applications can also be developed. More emotions and personality traits can be added. 3D artwork generation can also be implemented. Integration with AR/VR environments and cloud deployment can further enhance the system capabilities.

References

- Speech Emotion Recognition using Machine Learning – IEEE / Research Paper
  https://pmc.ncbi.nlm.nih.gov/articles/PMC10662716
- Librosa Python Library Documentation
  https://librosa.org/doc/latest/index.html
- TensorFlow Official Documentation
  https://www.tensorflow.org/
- Research Papers on Speech Emotion Recognition (SER)
  https://www.sciencedirect.com/science/article/pii/S2667305323000911
- Generative AI Research Articles
  https://arxiv.org/list/cs.AI/recent